CBSE Class 10 Maths Standard Question Paper 2023

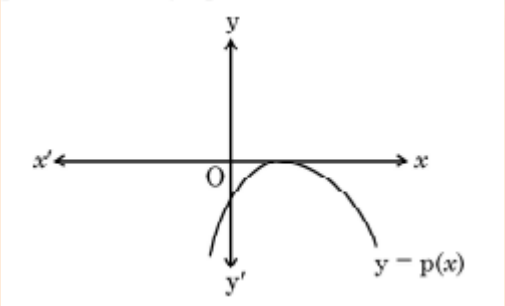
Set 1 (30/1/1) Solutions

SECTION - A

(Multiple Choice Questions)

Each question is of 1 mark.

1. The graph of $y = p(x)$ is given, for a polynomial $p(x)$. The number of zeroes of $p(x)$ from the graph is **1**



(A) 3

(B) 1

(C) 2

(D) 0

Answer: (A) 3

2. The value of k for which the pair of equations $kx = y + 2$ and $6x = 2y + 3$ has infinitely many solutions, **1**

(A) is $k = 3$

(B) does not exist

(C) is $k = -3$

(D) is $k = 4$

Answer: (A) is $k = 3$

3. If $p-1, p+1$ and $2p+3$ are in A.P., then the value of p is **1**

(A) -2

(B) 4

(C) 0

(D) 2

Answer: (C) 0

4. In what ratio, does x -axis divide the line segment joining the points $A(3,6)$ **and** $B(-12,-3)$ **?** **1**

(A) 1: 2

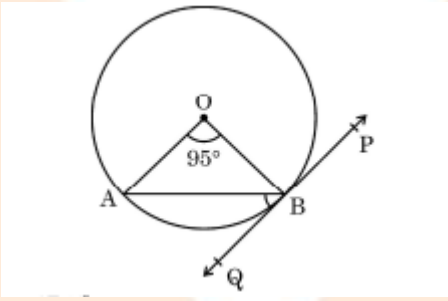
(B) **1: 4**

(C) **4: 1**

(D) **2: 1**

Answer: (D) **2: 1**

5. In the given figure, PQ is tangent to the circle centred at O . If $\angle AOB = 95^\circ$**, then the measure of** $\angle ABQ$ **will be** **1**



(A) 47.5°

(B) 42.5°

(C) 85°

(D) 95°

Answer: (B) 42.5°

6. If $2\tan A = 3$, then the value of $\frac{4\sin A + 3\cos A}{4\sin A - 3\cos A}$ is **1**

(A) $\frac{7}{\sqrt{13}}$

(B) $\frac{1}{\sqrt{13}}$

(C) 3

(D) does not exist

Answer: (C) 3

7. If α, β are the zeroes of a polynomial $p(x) = x^2 + x - 1$, then $\frac{1}{\alpha} + \frac{1}{\beta}$ equals to **1**

(A) 1

(B) 2

(C) -1

(D) $\frac{-1}{2}$

Answer: (A) 1

8. The least positive value of k , for which the quadratic equation $2x^2 + kx - 4 = 0$ has rational roots, is **1**

(A) $\pm 2\sqrt{2}$

(B) 2

(C) ± 2

(D) $\sqrt{2}$

Answer:

9. $\left[\frac{3}{4}\tan^2 30^\circ - \sec^2 45^\circ + \sin^2 60^\circ\right]$ **is equal to** **1**

(A) -1

(B) $\frac{5}{6}$

(C) $\frac{-3}{2}$

(D) $\frac{1}{6}$

Answer:(A) -1

10. Curved surface area of a cylinder of height 5 cm is $94.2\,\text{cm}^2$. Radius of the cylinder is (Take $\pi = 3.14$) **1**

(A) 2 cm

(B) 3 cm

(C) 2.9 cm

(D) 6 cm

Answer: (B) 3 cm

11. The distribution below gives the marks obtained by 80 students on a test: **1**

Marks	Less than 10	Less than 20	Less than 30	Less than 40	Less than 50	Less than 60
Number of Students	3	12	27	57	75	80

The modal class of this distribution is :

(A) $10-20$

(B) $20-30$

(C) $30-40$

(D) $50-60$

Answer: (C) $30-40$

12. The curved surface area of a cone having height 24 cm and radius 7 cm, is 1

(A) $528\,\text{cm}^2$

(B) $1056\,\text{cm}^2$

(C) $550\,\text{cm}^2$

(D) $500\,\text{cm}^2$

Answer: (C) $550\,\text{cm}^2$

13. The distance between the points $(0, 2\sqrt{5})$ **and** $(-2\sqrt{5}, 0)$ **is** **1**

(A) $2\sqrt{10}$ **units**

(B) $4\sqrt{10}$ **units**

(C) $2\sqrt{20}$ **units**

(D) 0

Answer:(A) $2\sqrt{10}$ units

14. Which of the following is a quadratic polynomial having zeroes $\frac{-2}{3}$ and $\frac{2}{3}$?1

(A) $4x^2 - 9$

(B) $\frac{4}{9}\left(9x^2 + 4\right)$

(C) $x^2 + \frac{9}{4}$

(D) $5\left(9x^2 - 4\right)$

Answer:(A) $4x^2 - 9$

15. If the value of each observation of a statistical data is increased by 3 , then the mean of the data 1

(A) remains unchanged

(B) increases by 3

(C) increases by 6

(D) increases by 3 n

Answer:(B) increases by 3

16. Probability of happening of an event is denoted by p and probability of non-happening of the event is denoted by q . Relation between p and q is 1

(A) $p+q=1$

(B) $p=1, q=1$

(C) $p=q-1$

(D) p+q+1=0

Answer: (A) $p+q=1$

17. A girl calculates that the probability of her winning the first prize in a lottery is 0.08. If 6000 tickets are sold, how many tickets has she bought? 1

(A) 40

(B) 240

(C) 480

(D) 750

Answer:(C) 480

18. In a group of 20 people, 5 can't swim. If one person is selected at random, then the probability that he/she can swim, is **1**

(A) $\frac{3}{4}$

(B) $\frac{1}{3}$

(C) 1

(D) $\frac{1}{4}$

Answer:(A) $\frac{3}{4}$

Assertion-Reason Type Questions

In Question 19 and 20, an Assertion (A) statement is followed by a statement of Reason (R). Select the correct option out of the following :

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(B) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).

(C) Assertion (A) is true but Reason (R) is false.

(D) Assertion (A) is false but Reason (R) is true.

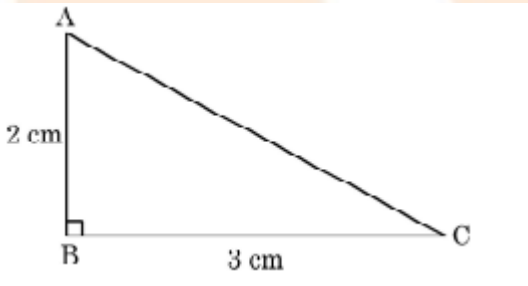
19. Assertion (A) : Point $P(0,2)$ is the point of intersection of y -axis with the line $x+2y=4$. **1**

Reason (R) : The distance of point $\mathrm{P}(0,2)$ from x -axis is 2 units.

Answer: (C) Assertion (A) is true but Reason (R) is false.

20. Assertion (A): The perimeter of $\triangle ABC$ is a rational number. **1**

Reason (R): The sum of the squares of two rational numbers is always rational.



Answer: (D) Assertion (A) is false but Reason (R) is true.

SECTION – B

(This section comprises of Very Short Answer (SA-I) type questions. Every question is of 2 marks.)

21.

(a) Solve the pair of equations $x=3$ and $y=-4$ graphically. **2**

Solution:

Plot the line $x=3$ on the graph. This line is a vertical line passing through $x=3$.

Plot the line $y=-4$ on the graph. This line is a horizontal line passing through $y=-4$.

The point of intersection of these lines is the solution, which is $(3,-4)$.

OR

(b) Using the graphical method, find whether the following system of linear equations is consistent or not: **2**

$x=0$ **and** $y=-7$**.**

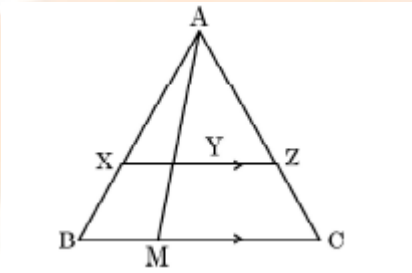
Solution:

Plot the line $x=0$ on the graph. This line is the y-axis.

Plot the line $y=-7$ on the graph. This line is a horizontal line passing through $y=-7$.

Since the lines intersect at a point, the system is consistent.

22.In the given figure, XZ is parallel to BC. $AZ = 3$ **cm,** $ZC = 2$ **cm,** $BM = 3$ **cm, and** $MC = 5$ **cm. Find the length of XY.** **2**



Solution:

By Basic Proportionality Theorem (Thales' Theorem):

$$\frac{AZ}{ZC} = \frac{BM}{MC} = \frac{XY}{YC}$$

Substituting the given values:

$$\frac{3}{2} = \frac{3}{5} = \frac{XY}{YC}$$

Cross multiply to find XY:

$$2XY = 3YC \quad \Rightarrow \quad XY = \frac{3}{2}YC$$

Let $XY = x$ and $YC = y$:

$$x = \frac{3}{2}y$$

Since $YC = ZC = 2$ cm:

$x = \frac{3}{2} \times 2 = 3$ cm

23.

(a) If $\sin\theta + \cos\theta = \sqrt{3}$, then find the value of $\sin\theta \cdot \cos\theta$. **2**

Solution:

Square both sides of the given equation:

$(\sin\theta + \cos\theta)^2 = (\sqrt{3})^2 \quad \Rightarrow \quad \sin^2\theta + \cos^2\theta + 2\sin\theta\cos\theta = 3$

Use $\sin^2\theta + \cos^2\theta = 1$:

$1 + 2\sin\theta\cos\theta = 3 \quad \Rightarrow \quad 2\sin\theta\cos\theta = 2 \quad \Rightarrow \quad \sin\theta\cos\theta = 1$

OR

(b) If $\sin\alpha = \frac{1}{\sqrt{2}}$ and $\cot\beta = \sqrt{3}$, then find the value of $\csc\alpha + \csc\beta$. **2**

Solution:

Given $\sin\alpha = \frac{1}{\sqrt{2}}$, hence $\csc\alpha = \frac{1}{\sin\alpha} = \sqrt{2}$.

Given $\cot\beta = \sqrt{3}$, so $\tan\beta = \frac{1}{\sqrt{3}}$. Thus, $\sin\beta = \frac{1}{2}$ and $\csc\beta = \frac{1}{\sin\beta} = 2$.

Therefore, $\csc\alpha + \csc\beta = \sqrt{2} + 2$.

24.Find the greatest number which divides 85 and 72 leaving remainders 1 and 2 respectively. 2

Solution:

Let the required number be x. Then,

$85-1=84$ and $72-2=70$

The required number x divides both 84 and 70. The greatest common divisor (GCD) of 84 and 70 is 14.

Hence, the required number is $x=14$.

25.A bag contains 4 red, 3 blue, and 2 yellow balls. One ball is drawn at random from the bag. Find the probability that the drawn ball is 2

(i)Red.

Solution:

Total number of balls = 4 + 3 + 2 = 9.

Number of red balls = 4.

Probability of drawing a red ball:

$$P(\text{Red})=\frac{4}{9}$$

(ii)Yellow.

Solution:

Number of yellow balls = 2.

Probability of drawing a yellow ball:

$P(\text{Yellow}) = \frac{2}{9}$.

SECTION – C

This section comprises of Short Answer (SA-II) type questions of 3 marks each.

26.Half of the difference between two numbers is 2. The sum of the greater number and twice the smaller number is 13. Find the numbers. 3

Solution:

Let the greater number be x and the smaller number be y.

According to the problem:

$\frac{x-y}{2} = 2 \quad \Rightarrow \quad x - y = 4$ (Equation 1)

The sum of the greater number and twice the smaller number:

$x + 2y = 13$ (Equation 2)

Substitute $x = y + 4$ (from Equation 1) into Equation 2:

$(y+4)+2y=13 \quad \Rightarrow \quad 3y+4=13 \quad \Rightarrow \quad 3y=9 \quad \Rightarrow \quad y=3$

Substitute $y=3$ back into Equation 1

$x-3=4 \quad \Rightarrow \quad x=7$

Answer: The numbers are 7 and 3.

27.Prove that $\sqrt{5}$ is an irrational number. **3**

Solution:

Assume that $\sqrt{5}$ is a rational number.

Then it can be written as $\sqrt{5}=\frac{a}{b}$, where a and b are integers with no common factors.

Squaring both sides, we get:

$5=\frac{a^2}{b^2} \quad \Rightarrow \quad a^2=5b^2$

This implies that a^2 is divisible by 5, so a must also be divisible by 5. Let $a=5k$.

Substituting $a=5k$ into $a^2=5b^2$, we get:

$(5k)^2=5b^2 \quad \Rightarrow \quad 25k^2=5b^2 \quad \Rightarrow \quad b^2=5k^2$

This implies that b^2 is divisible by 5, so b must also be divisible by 5.

But this contradicts the assumption that a and b have no common factors.

Therefore, $\sqrt{5}$ is irrational.

28.If $(-5,3)$ and $(5,3)$ are two vertices of an equilateral triangle, then find coordinates of the third vertex, given that origin lies inside the triangle. (Take $\sqrt{3}=1.7$) **3**

Solution:

Let the vertices be $A(-5,3)$, $B(5,3)$, and $C(x,y)$.

The distance between A and B is 10 units (since they lie on a horizontal line).

Since ABC is an equilateral triangle, $AB=BC=AC=10$ units.

Using the distance formula between $A(-5,3)$ and $C(x,y)$:

$$\sqrt{(x+5)^2+(y-3)^2}=10$$

Simplifying:

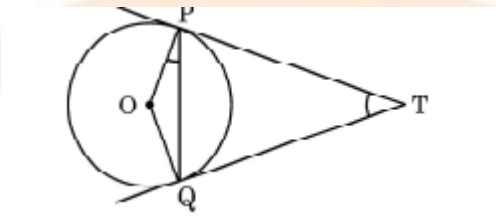
$$(x+5)^2+(y-3)^2=100$$

Similarly, for $B(5,3)$ and $C(x,y)$:

$$\sqrt{(x-5)^2+(y-3)^2}=10 \quad \Rightarrow \quad (x-5)^2+(y-3)^2=100$$

Solving these two equations gives the coordinates of C as $(0,3+5\sqrt{3})$ or $(0,3-5\sqrt{3})$.

29. (a)Two tangents TP and TQ are drawn to a circle with center O from an external point T. Prove that $\angle PTQ = 2\angle OPQ$. **3**



Solution:

Given that TP and TQ are tangents to the circle from point T, we know that the tangents from an external point to a circle are equal in length. Therefore, $TP = TQ$.

Since the radii drawn to the tangents at points P and Q are perpendicular to the tangents, we have $\angle OPT = \angle OQT = 90^\circ$.

In the quadrilateral TPQO, the sum of the angles is 360°:

$\angle PTQ + \angle OPT + \angle OQT + \angle POQ = 360^\circ$

Substituting the known angles:

$\angle PTQ + 90^\circ + 90^\circ + \angle POQ = 360^\circ$

$\angle PTQ + \angle POQ = 180^\circ$

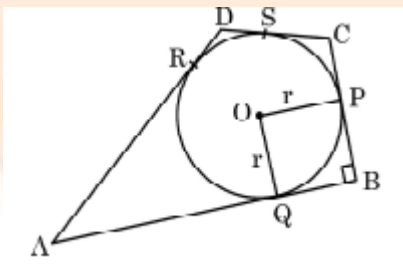
Now, since $\angle POQ$ subtends arc PQ, it follows that:

$\angle POQ = 2 \times \angle OPQ$

Thus, $\angle PTQ = 2 \times \angle OPQ$.

OR

(b)In the given figure, a circle is inscribed in a quadrilateral ABCD in which $\angle B = 90^\circ$. If $AD = 17$ cm, $AB = 20$ cm, and $DS = 3$ cm, then find the radius of the circle. 3



Solution:

Given that the circle is inscribed in quadrilateral ABCD and $\angle B = 90^\circ$, we have $AB + CD = AD + BC$. Let the radius of the circle be r.

The area of quadrilateral ABCD can be given by:

$$\text{Area} = \text{semi-perimeter} \times r$$

Calculate the semi-perimeter:

$$s = \frac{AB + BC + CD + DA}{2} = \frac{20 + BC + 17 + 3}{2} = 20 + BC$$

We know the area of the quadrilateral using the Pythagorean Theorem:

$$\text{Area} = AB \times BC + \frac{1}{2} \times r \times (AB + AD + DA + CD)$$

30.Prove that: $\frac{\tan\theta+\sec\theta-1}{\tan\theta-\sec\theta+1}=\frac{1+\sin\theta}{\cos\theta}$ **3**

Solution:

Express $\tan\theta$ and $\sec\theta$ in terms of $\sin\theta$ and $\cos\theta$: $\tan\theta=\frac{\sin\theta}{\cos\theta}$, $\sec\theta=\frac{1}{\cos\theta}$

Substitute into the equation: $\frac{\frac{\sin\theta}{\cos\theta}+\frac{1}{\cos\theta}-1}{\frac{\sin\theta}{\cos\theta}-\frac{1}{\cos\theta}+1}=\frac{\frac{\sin\theta+1-\cos\theta}{\cos\theta}}{\frac{\sin\theta-1+\cos\theta}{\cos\theta}}=\frac{\sin\theta+1-\cos\theta}{\sin\theta-1+\cos\theta}$

Simplifying the numerator and denominator, we arrive at: $\frac{1+\sin\theta}{\cos\theta}$

31.

(a)A room is in the form of a cylinder surmounted by a hemispherical dome. The base radius of the hemisphere is one-half the height of the cylindrical part. Find the total height of the room if it contains $\frac{1408}{21}$ m^3 of air. $(Take\pi=\frac{22}{7})$ 3

Solution:

Let the radius of the base of the hemisphere be r. Then, the height of the cylindrical part is 2r.

The volume of the air in the room is the sum of the volumes of the cylinder and the hemisphere:

$$V=\pi r^2h+\frac{2}{3}\pi r^3$$

where h is the height of the cylinder.

Substituting the values:

$$\frac{1408}{21} = \pi r^2(2r) + \frac{2}{3}\pi r^3 = \frac{22}{7} \times r^3\left(2 + \frac{2}{3}\right)$$

Simplifying:

$$\frac{1408}{21} = \frac{22}{7} \times r^3 \times \frac{8}{3}$$

Solving for r^3:

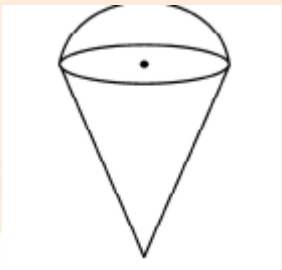
$$r^3 = \frac{1408 \times 3}{21 \times 8} = \frac{4224}{168} = 25.14 \quad \Rightarrow \quad r \approx 2.92 \text{ m}$$

Therefore, the height of the room (cylinder + hemisphere) is $2r + r = 3r \approx 3 \times 2.92 = 8.76$ m
.

Or

(b)An empty cone is of radius 3 cm and height 12 cm. Ice cream is filled in it so that the lower part of the cone, which is $\left(\frac{1}{6}\right)^{th}$ of the volume of the cone, is unfilled, but a hemisphere is formed on the top. Find the volume of the ice cream. $(Take \pi = 3.14)$

3



Solution:

The volume of the cone is given by:

$$V_{cone} = \frac{1}{3}\pi r^2 h = \frac{1}{3} \times 3.14 \times 3^2 \times 12 = 113.04 \text{ cm}^3$$

The unfilled portion is $\frac{1}{6}$ of the volume:

$$V_{unfilled} = \frac{1}{6} \times 113.04 = 18.84 \text{ cm}^3$$

Therefore, the volume of the filled cone part:

$$V_{filled} = 113.04 - 18.84 = 94.2 \text{ cm}^3$$

The volume of the hemisphere:

$$V_{hemisphere} = \frac{2}{3}\pi r^3 = \frac{2}{3} \times 3.14 \times 3^3 = 56.52 \text{ cm}^3$$

Total volume of the ice cream:

$V_{\text{ice cream}} = V_{\text{filled}} + V_{\text{hemisphere}} = 94.2 + 56.52 = 150.72 \text{ cm}^3$.

SECTION – D

This section comprises of Long Answer (LA) type questions of 5 marks each.

32.If a line is drawn parallel to one side of a triangle to intersect the other two sides at distinct points, prove that the other two sides are divided in the same ratio. 5

Solution:

itemize

Consider a triangle $\triangle ABC$ with DE drawn parallel to BC intersecting AB at D and AC at E.

We need to prove that $\frac{AD}{DB} = \frac{AE}{EC}$.

By the Basic Proportionality Theorem (Thales theorem), if a line is drawn parallel to one side of a triangle to intersect the other two sides, then the other two sides are divided in the same ratio.

Hence, $\frac{AD}{DB} = \frac{AE}{EC}$.

33.

(a) The angle of elevation of the top of a tower 24 m high from the foot of another tower in the same plane is 60°. The angle of elevation of the top of the second tower from the foot of the first tower is 30°. Find the distance between two towers and the height of the other tower. Also, find the length of the wire attached to the tops of both the towers. 5

Solution:

Let the height of the first tower be 24 m and the height of the second tower be h m. Let the distance between the towers be d.

Using trigonometry in the triangle formed by the second tower:

$$\tan 30^\circ = \frac{h}{d} \quad \Rightarrow \quad \frac{1}{\sqrt{3}} = \frac{h}{d} \quad \Rightarrow \quad h = \frac{d}{\sqrt{3}}$$

Using trigonometry in the triangle formed by the first tower:

$$\tan 60^\circ = \frac{24}{d} \quad \Rightarrow \quad \sqrt{3} = \frac{24}{d} \quad \Rightarrow \quad d = \frac{24}{\sqrt{3}} = 8\sqrt{3} \text{ m}$$

Therefore, $h = \frac{8\sqrt{3}}{\sqrt{3}} = 8$ m.

The length of the wire attached to the tops of both the towers is:

$$\text{Length of wire} = \sqrt{(d)^2 + (h-24)^2} = \sqrt{(8\sqrt{3})^2 + (8-24)^2} = \sqrt{192+256} = \sqrt{448} \approx 21.17 \text{ m}$$

OR

(b) A spherical balloon of radius r subtends an angle of 60° at the eye of an observer. If the angle of elevation of its centre is 45° from the same point, prove that height of the centre of the balloon is $\sqrt{2}$ times its radius. 5

Solution:

Let the height of the centre of the balloon from the observer's eye be h.

From the given condition, $\tan 45^\circ = \frac{h}{r}$, hence $h = r$.

For the given subtended angle 60°, the length of the arc subtending this angle is:

$$r\theta = r \times \frac{\pi}{3}$$

By Pythagoras theorem:

$$\text{Length of arc} = \sqrt{(h+r)^2 - r^2} = \sqrt{2r^2} = \sqrt{2}r$$

Therefore, height of the centre of the balloon is $\sqrt{2}$ times its radius.

34.A chord of a circle of radius 14 cm subtends an angle of 60° at the centre. Find the area of the corresponding minor segment of the circle. Also, find the area of the major segment of the circle. 5

Solution:

The area of the sector:

$$\text{Area of sector} = \frac{\theta}{360^\circ} \times \pi r^2 = \frac{60^\circ}{360^\circ} \times \frac{22}{7} \times 14^2 = \frac{1}{6} \times \frac{22}{7} \times 196 = 44.59 \text{ square cm}$$

The area of the triangle formed by the chord:

Area of triangle $= \frac{1}{2} r^2 \sin\theta = \frac{1}{2} \times 14 \times 14 \times \sin 60^\circ = \frac{1}{2} \times 196 \times \frac{\sqrt{3}}{2} = 84.87$ square cm

The area of the minor segment:

Area of minor segment = Area of sector – Area of triangle = $44.59 - 84.87 = -40.28$ square cm

The area of the major segment:

Area of major segment $= \pi r^2 -$ Area of minor segment $= \frac{22}{7} \times 14^2 - (-40.28) = 616 + 40.28 = 656.28$ square cm

.

35.

(a) The ratio of the 11th term to 17th term of an A.P. is 3:4. Find the ratio of 5th term to 21st term of the same A.P. Also, find the ratio of the sum of first 5 terms to that of first 21 terms. 5

Solution:

Let the first term of the A.P. be a and the common difference be d.

Given:

$$\frac{a+10d}{a+16d} = \frac{3}{4}$$

We can express this as:

$$4(a+10d) = 3(a+16d) \quad \Rightarrow \quad 4a+40d = 3a+48d \quad \Rightarrow \quad a = 8d$$

The ratio of the 5th term to 21st term:

$$\frac{a+4d}{a+20d} = \frac{8d+4d}{8d+20d} = \frac{12d}{28d} = \frac{3}{7}$$

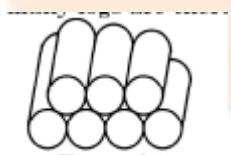
The ratio of the sum of first 5 terms to that of first 21 terms:

$$\frac{S_5}{S_{21}} = \frac{\frac{5}{2}\times[2a+4d]}{\frac{21}{2}\times[2a+20d]} = \frac{5\times(2a+4d)}{21\times(2a+20d)} = \frac{5\times(16d)}{21\times(48d)} = \frac{5}{21}\times\frac{16}{48} = \frac{5}{63}.$$

OR

(b) 250 logs are stacked in the following manner: 22 logs in the bottom row, 21 in the next row, 20 in the row next to it and so on (as shown by an example). In how many rows are the 250 logs placed and how many logs are there in the top row?

5



(Example)

Solution:

This is an arithmetic progression with first term $a = 22$, common difference $d = -1$, and total logs $S_n = 250$.

The sum of the first n terms is given by:

$$S_n = \frac{n}{2}\times[2a+(n-1)d] = \frac{n}{2}\times[44-n+1] = \frac{n}{2}\times[45-n] = 250$$

Simplifying:

$$n(45-n) = 500 \quad \Rightarrow \quad n^2 - 45n + 500 = 0$$

Solving the quadratic equation:

$$n=\frac{45\pm\sqrt{45^2-4\times1\times500}}{2}=\frac{45\pm\sqrt{2025-2000}}{2}=\frac{45\pm\sqrt{25}}{2}=\frac{45\pm5}{2}$$

$$n=\frac{50}{2}=25 \text{ or } n=\frac{40}{2}=20$$

Since $n=20$, the top row will have $22-19=3$ logs.

SECTION – E

In this section, there are 3 case study/passage-based questions. Each question is of 4 marks.

Case Study :

36. While designing the school year book, a teacher asked the student that the length and width of a particular photo is increased by x units each to double the area of the photo. The original photo is 18 cm long and 12 cm wide.

Based on the above information, answer the following questions:

(I)Write an algebraic equation depicting the above information. 1

Solution:

The original area of the photo is $18\times12=216$ square cm .

The new dimensions are $(18+x)$ and $(12+x)$.

Since the area is doubled:

$(18+x)(12+x)=2\times216=432$

(II)Write the corresponding quadratic equation in standard form. **1**

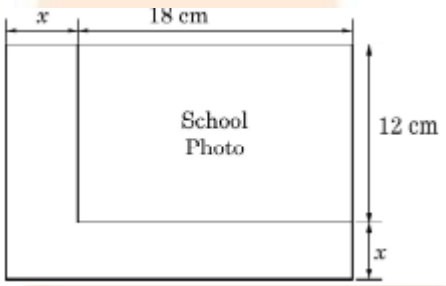
Solution:

Expanding the equation:

$$18\times12+30x+x^2=432 \quad \Rightarrow \quad x^2+30x-216=0$$

(III)What should be the new dimensions of the enlarged photo? **2**

Can any rational value of x make the new area equal to 220 cm^2



Solution:

Solving the quadratic equation using the quadratic formula:

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}=\frac{-30\pm\sqrt{30^2+864}}{2}$$

This gives $x=6$ or $x=-36$. Discard the negative value.

The new dimensions are $18+6=24$ cm and $12+6=18$ cm.

37.India Meteorological Department observes seasonal and annual rainfall every year in different sub-divisions of our country. 4

It helps them to compare and analyse the results. The table given below shows sub-division wise seasonal (monsoon) rainfall (mm) in 2018:

Rainfall (mm)	Number of Sub-divisions
200-400	2
400-600	4
600-800	7
800-1000	4
1000-1200	2
1200-1400	3
1400-1600	1
1600-1800	1

Based on the above information, answer the following questions

(I)Write the modal class. 1

Solution: The modal class is the class interval with the highest frequency. Here, the modal class is 600-800 mm, as it has the highest frequency of 7 sub-divisions.

(II)Find the median of the given data. **1**

Solution:

The total number of sub-divisions is $2+4+7+4+2+3+1+1=24$.

The median class is the class where the cumulative frequency crosses $\frac{N}{2}=12$.

Cumulative frequency:

Class Interval	Frequency	Cumulative Frequency
200 – 400	2	2
400 – 600	4	6
600 – 800	7	13
800 – 1000	4	17
1000 – 1200	2	19
1200 – 1400	3	22
1400 – 1600	1	23
1600 – 1800	1	24

The median is calculated using the formula:

$$\text{Median} = L + \left(\frac{\frac{N}{2} - F}{f} \right) \times h$$

where $L = 600$, $N = 24$, $F = 6$, $f = 7$, and $h = 200$.

Substituting the values:

$$\text{Median} = 600 + \left(\frac{12-6}{7} \right) \times 200 = 600 + \left(\frac{6}{7} \right) \times 200 = 600 + 171.43 = 771.43 \text{ mm}$$

OR

(III)Find the mean rainfall in this season. **2**

Solution:

The mean rainfall is calculated by finding the sum of midpoints multiplied by frequencies and then dividing by the total frequency.

Midpoints are: 300, 500, 700, 900, 1100, 1300, 1500, 1700.

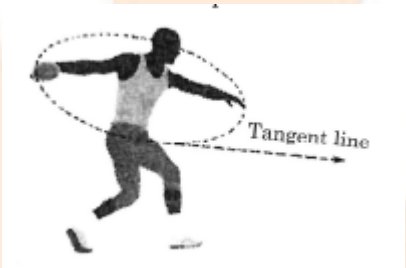
Mean

$$\text{Mean} = \frac{2\times300 + 4\times500 + 7\times700 + 4\times900 + 2\times1100 + 3\times1300 + 1\times1500 + 1\times1700}{24} = \frac{38700}{24} = 1612.5 \text{ mm}$$

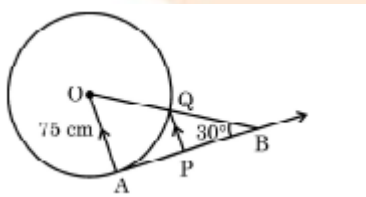
(III)If a sub-division having at least 1000 mm rainfall during the monsoon season is considered a good rainfall sub-division, then how many sub-divisions had good rainfall? 2

Solution: The sub-divisions with good rainfall are those in the intervals 1000-1200, 1200-1400, 1400-1600, and 1600-1800. Adding these frequencies: $2+3+1+1=7$ sub-divisions had good rainfall.

38.The discus throw is an event in which an athlete attempts to throw a discus. The athlete spins anti-clockwise around one and a half times through a circle, then releases the throw. When released, the discus travels along the tangent to the circular spin orbit.



In the given figure, AB is one such tangent to a circle of radius 75 cm. Point O is the centre of the circle and $\angle ABO = 30^{\circ}$. PQ is parallel to OA.



Based on the above information:

(a) Find the length ofAB. **1**

Solution:

We know that the length of the tangent from a point outside the circle is given by:

$AB = \sqrt{OB^2 - OA^2}$

Here, $OB = 75$ cm (radius) and $\angle ABO = 30^\circ$.

Using the cosine rule in $\triangle ABO$

$AB = \sqrt{OB^2 - OA^2 \cos^2 \angle ABO}$

Substituting the values:

$$AB = \sqrt{75^2 - 75^2 \cos^2 30^\circ} = \sqrt{5625 - 5625 \times \frac{3}{4}} = \sqrt{5625 - 4218.75} = \sqrt{1406.25} \approx 37.5 \text{ cm}$$

(b)Find the length of OB. **1**

Solution: The length ofOBis the radius of the circle, which is 75 cm.

(c) Find the length ofAP **2**

Solution:

Since PQ is parallel to OA, APis a straight line.

item Therefore,AP = AB + BP. We already know $AB = 37.5$ cm.

BP can be found by:

$BP = AB \times \tan 30^{\circ} = 37.5 \times \frac{1}{\sqrt{3}} \approx 21.65$ cm

So, $AP = 37.5 + 21.65 = 59.15$ cm.

OR

Find the length of PQ. **2**

Solution:

Since PQ is parallel to OA, PQ can be considered as another tangent. Thus, its length will be the same as AB:

$PQ = AB = 37.5$ cm